

PRAKTIKUM BASIS DATA

MODUL 8

Scripting SQL Create



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

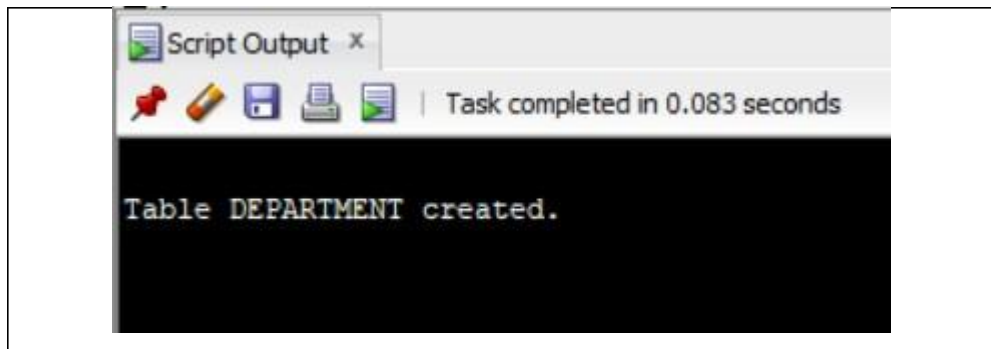
Jurnal

1. Buatlah desain database untuk suatu perusahaan:
 - a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.

Query

```
CREATE TABLE department (  
  department_id NUMBER(4) PRIMARY KEY,  
  department_name VARCHAR2(100),  
  manager_id NUMBER(6),  
  location_id NUMBER(4)  
);
```

Screenshot

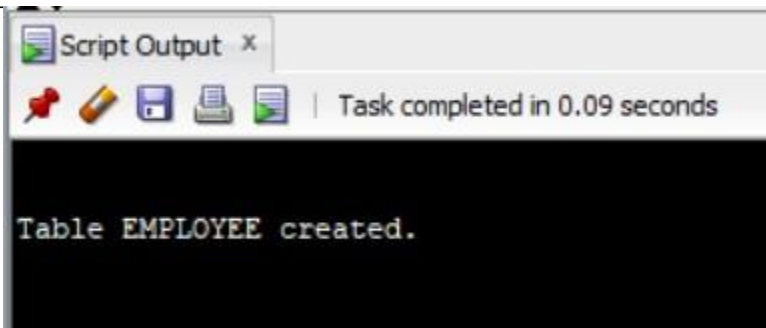


- b. Table employee dengan constraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_name NOT NULL, email, salary, commission_pct, hire_date NOT NULL.

Query

```
CREATE TABLE employee (  
  employee_id NUMBER(6) PRIMARY KEY,  
  last_name VARCHAR2(100) NOT NULL,  
  email VARCHAR2(100),  
  salary NUMBER(10,2),  
  commission_pct NUMBER(4,2),  
  hire_date DATE NOT NULL,  
  department_id NUMBER(4),  
  CONSTRAINT emp_dept_fk FOREIGN KEY (department_id)  
  REFERENCES department(department_id)  
);
```

Screenshot

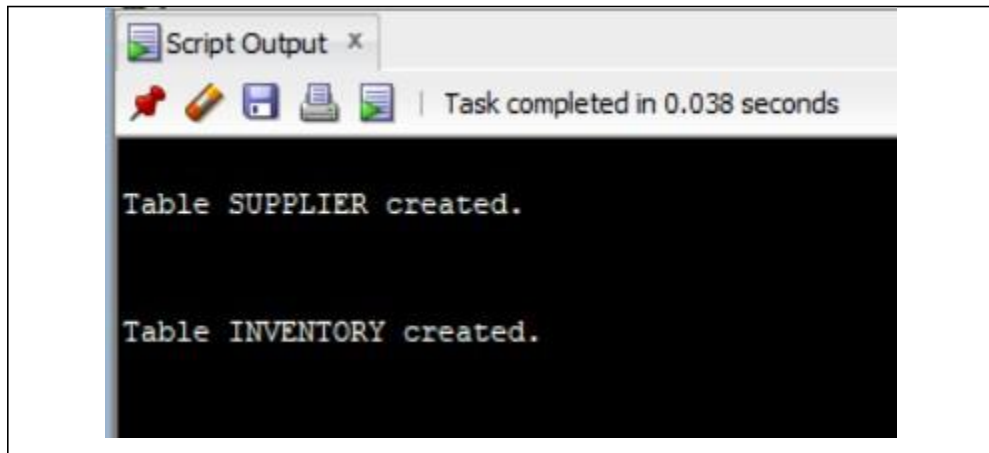


- c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).

Query

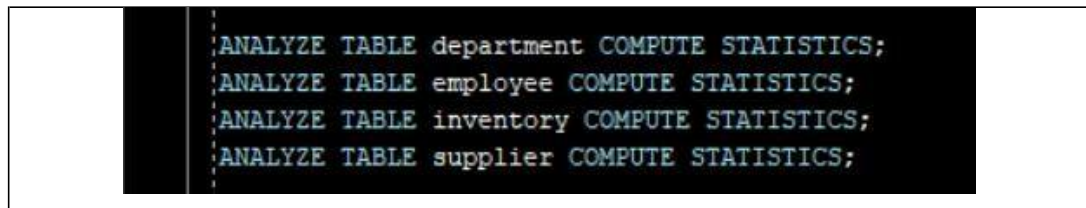
```
CREATE TABLE supplier (  
  supplier_id NUMBER(4) PRIMARY KEY,  
  supplier_name VARCHAR2(100),  
  contact_name VARCHAR2(100),  
  phone_number VARCHAR2(20)  
);  
  
CREATE TABLE inventory (  
  inventory_id NUMBER(4) PRIMARY KEY,  
  item_name VARCHAR2(100),  
  quantity NUMBER(6),  
  price NUMBER(10,2),  
  supplier_id NUMBER(4),  
  CONSTRAINT inv_supplier_fk FOREIGN KEY (supplier_id)  
  REFERENCES supplier(supplier_id)  
);
```

Screenshot

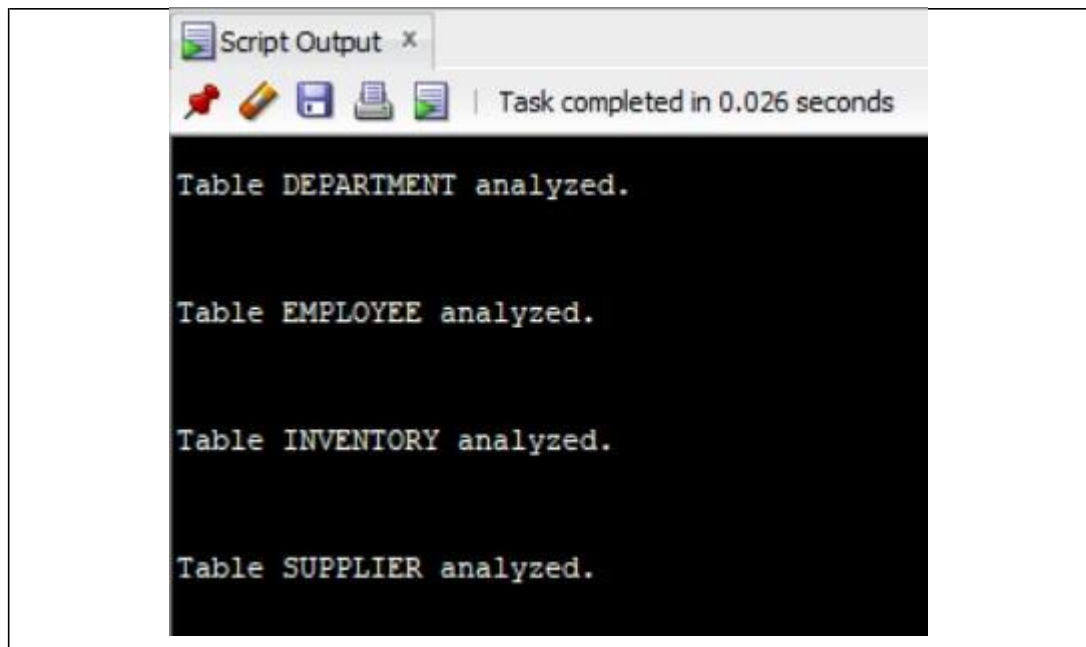


2. Analisis table-table dengan compute statistics

Query



Screenshot



3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).

Query

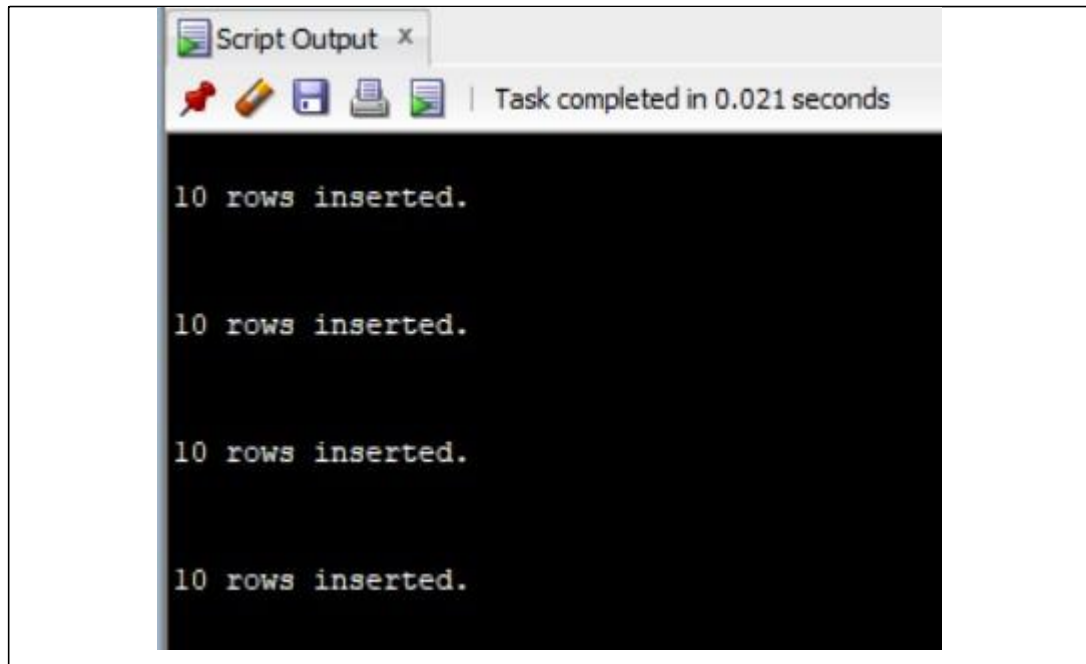
```
INSERT ALL
  INTO department VALUES (10, 'HR', 101, 1)
  INTO department VALUES (20, 'Finance', 102, 2)
  INTO department VALUES (30, 'IT', 103, 3)
  INTO department VALUES (40, 'Sales', 104, 4)
  INTO department VALUES (50, 'Marketing', 105, 5)
  INTO department VALUES (60, 'R and D', 106, 6)
  INTO department VALUES (70, 'Legal', 107, 7)
  INTO department VALUES (80, 'Customer Service', 108, 8)
  INTO department VALUES (90, 'Operations', 109, 9)
  INTO department VALUES (100, 'Logistics', 110, 10)
SELECT * FROM dual;

INSERT ALL
  INTO employee VALUES (1, 'Smith', 'john.smith@example.com', 50000, 0.10, DATE '2020-01-01', 10)
  INTO employee VALUES (2, 'Johnson', 'jane.johnson@example.com', 60000, 0.15, DATE '2019-02-01', 20)
  INTO employee VALUES (3, 'Williams', 'mark.williams@example.com', 70000, 0.20, DATE '2018-03-01', 30)
  INTO employee VALUES (4, 'Brown', 'lisa.brown@example.com', 80000, 0.25, DATE '2017-04-01', 40)
  INTO employee VALUES (5, 'Davis', 'paul.davis@example.com', 55000, 0.18, DATE '2021-05-01', 50)
  INTO employee VALUES (6, 'Miller', 'anna.miller@example.com', 65000, 0.22, DATE '2020-06-01', 60)
  INTO employee VALUES (7, 'Wilson', 'susan.wilson@example.com', 75000, 0.30, DATE '2019-07-01', 70)
  INTO employee VALUES (8, 'Moore', 'david.moore@example.com', 85000, 0.35, DATE '2018-08-01', 80)
  INTO employee VALUES (9, 'Taylor', 'luke.taylor@example.com', 95000, 0.40, DATE '2017-09-01', 90)
  INTO employee VALUES (10, 'Anderson', 'rachel.anderson@example.com', 45000, 0.05, DATE '2022-10-01', 100)
SELECT * FROM dual;
```

```
INSERT ALL
  INTO supplier VALUES (1, 'Tech Supplier Inc.', 'John Doe', '1234567890')
  INTO supplier VALUES (2, 'Office Supplies Ltd.', 'Jane Smith', '9876543210')
  INTO supplier VALUES (3, 'Gadget Store', 'Mark Brown', '5555555555')
  INTO supplier VALUES (4, 'Electronic Goods Co.', 'Sarah White', '4444444444')
  INTO supplier VALUES (5, 'Print Solutions', 'Paul Green', '3333333333')
  INTO supplier VALUES (6, 'Digital Source', 'Kevin Lee', '2222222222')
  INTO supplier VALUES (7, 'Smart Devices', 'Lisa Ray', '1111111111')
  INTO supplier VALUES (8, 'Computer Hub', 'Andi Putra', '6666666666')
  INTO supplier VALUES (9, 'Office Mart', 'Sinta Dewi', '7777777777')
  INTO supplier VALUES (10, 'Global Elektronik', 'Rudi Harto', '8888888888')
SELECT * FROM dual;

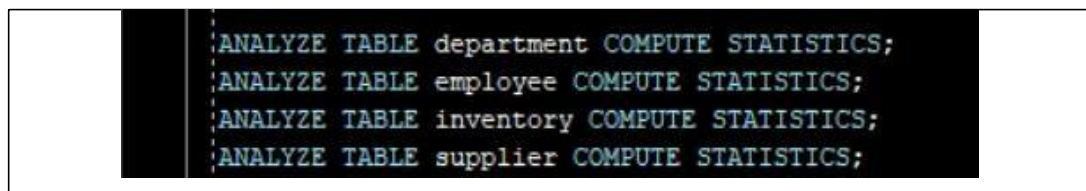
INSERT ALL
  INTO inventory VALUES (1, 'Laptop', 20, 10000000, 1)
  INTO inventory VALUES (2, 'Mouse', 50, 150000, 2)
  INTO inventory VALUES (3, 'Keyboard', 30, 250000, 3)
  INTO inventory VALUES (4, 'Monitor', 10, 2000000, 4)
  INTO inventory VALUES (5, 'Printer', 15, 1500000, 5)
  INTO inventory VALUES (6, 'Hard Drive', 40, 750000, 6)
  INTO inventory VALUES (7, 'Webcam', 25, 500000, 7)
  INTO inventory VALUES (8, 'Speakers', 35, 600000, 8)
  INTO inventory VALUES (9, 'USB Cable', 100, 50000, 9)
  INTO inventory VALUES (10, 'Phone', 10, 3000000, 10)
SELECT * FROM dual;
```

Screenshot

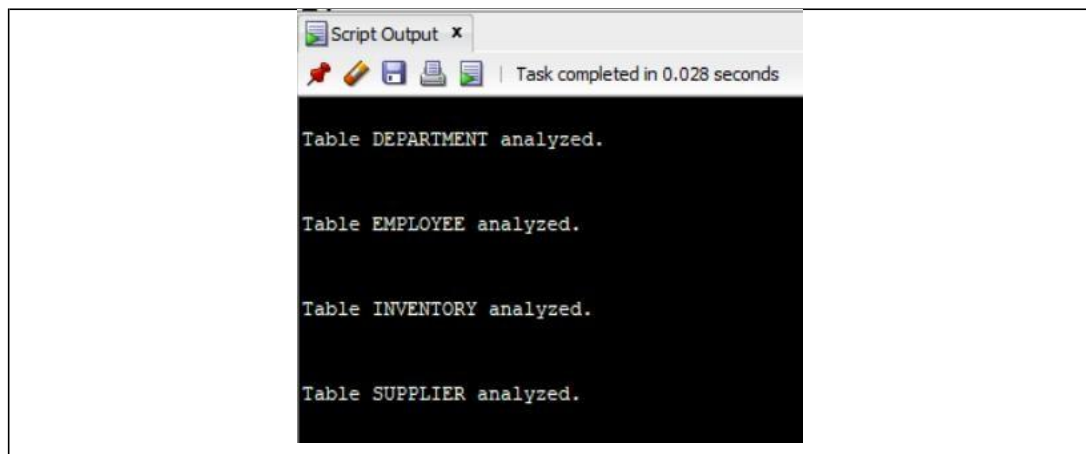


4. Analisis table-table dengan compute statistics

Query



Screenshot

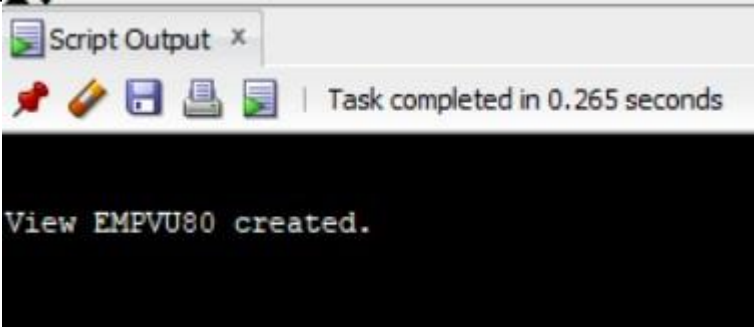


5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.

Query

```
CREATE OR REPLACE VIEW empvu80 AS
SELECT
employee_id AS id_number,
last_name   AS name,
salary,
department_id
FROM employee
WHERE department_id = 80;
```

Screenshot



Script Output x

Task completed in 0.265 seconds

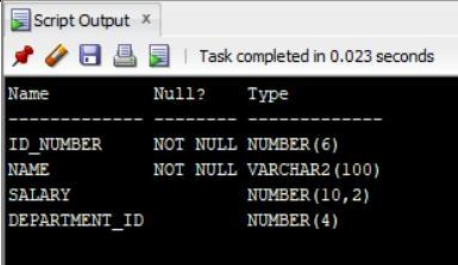
View EMPVU80 created.

6. Kemudian tampilkan struktur dari view empvu80.

Query

```
DESC empvu80;
```

Screenshot



Script Output x

Task completed in 0.023 seconds

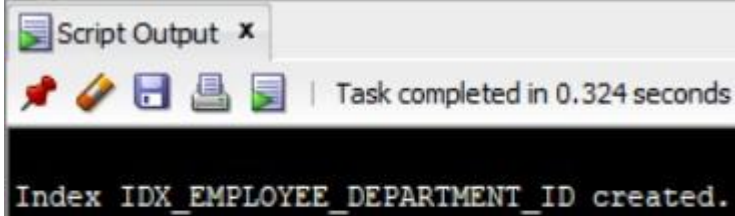
Name	Null?	Type
ID_NUMBER	NOT NULL	NUMBER(6)
NAME	NOT NULL	VARCHAR2(100)
SALARY		NUMBER(10,2)
DEPARTMENT_ID		NUMBER(4)

7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

Query

```
CREATE INDEX idx_employee_department_id  
ON employee(department_id);
```

Screenshot



The screenshot shows a 'Script Output' window with a toolbar containing icons for a pin, pencil, save, print, and refresh. The text in the window reads: 'Task completed in 0.324 seconds' followed by 'Index IDX_EMPLOYEE_DEPARTMENT_ID created.' on a black background with white text.